

WDI 2022 Report

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Table of contents

Test Chunk	1
Data overview	1
Indicators selected	2
Summary table	3
Results	6

Test Chunk

```
import pandas as pd
df = pd.read_csv("wdi.csv")
df.head()
```

	country	inflation_rate	exports_gdp_share	gdp_growth_rate	gdp_per_capita	adult_literacy
0	Afghanistan	NaN	18.380042	-6.240172	352.603733	NaN
1	Albania	6.725203	37.395422	4.856402	6810.114041	98.5
2	Algeria	9.265516	31.446856	3.600000	5023.252932	NaN
3	American Samoa	NaN	46.957520	1.735016	19673.390102	NaN
4	Andorra	NaN	NaN	9.563798	42350.697069	NaN

Data overview

```
list(df.columns)
```

```
[ 'country',
  'inflation_rate',
  'exports_gdp_share',
  'gdp_growth_rate',
  'gdp_per_capita',
  'adult_literacy_rate',
  'primary_school_enrolment_rate',
  'education_expenditure_gdp_share',
  'measles_immunisation_rate',
  'health_expenditure_gdp_share',
  'income_inequality',
  'unemployment_rate',
  'life_expectancy',
  'total_population']
```

Indicators selected

In this report, I focus on three indicators: - Inflation rate (`inflation_rate`) - GDP per capita (`gdp_per_capita`) - Life expectancy (`life_expectancy`)

```
country_col = "country"
ind1 = "inflation_rate"
ind2 = "gdp_per_capita"
ind3 = "life_expectancy"
```

```
df[ind1].describe()
```

```
count      169.000000
mean        12.493936
std         19.682433
min         -6.687321
25%          5.518129
50%          7.967574
75%         11.665567
max         171.205491
Name: inflation_rate, dtype: float64
```

The inflation rate is available for 169 countries. The mean inflation rate is approximately 12.49%, while the median is much lower at 7.97%, suggesting the presence of extreme outliers. The maximum value reaches 171.21%, indicating that at least one country experienced extremely high inflation in 2022. The large standard deviation (19.68) further confirms substantial

variation across countries. The negative minimum value (-6.69%) suggests that some countries experienced deflation.

```
df[ind2].describe()
```

```
count      203.000000
mean       20345.707649
std        31308.942225
min         259.025031
25%        2570.563284
50%        7587.588173
75%        25982.630050
max        240862.182448
Name: gdp_per_capita, dtype: float64
```

GDP per capita is available for 203 countries. The mean value is 20,345 USD, while the median is significantly lower at 7,588 USD, indicating a strongly right-skewed distribution. The maximum GDP per capita reaches 240,862 USD, reflecting a few very high-income economies that pull the mean upward. The large standard deviation (31,309) further demonstrates substantial inequality in income levels across countries.

```
df[ind3].describe()
```

```
count      209.000000
mean        72.416519
std          7.713322
min         52.997000
25%         66.782000
50%         73.514634
75%         78.475000
max         85.377000
Name: life_expectancy, dtype: float64
```

Life expectancy data is available for 209 countries. The mean life expectancy is 72.42 years, and the median is 73.51 years, which are very close, suggesting a relatively symmetric distribution. The range spans from 52.99 years to 85.38 years, indicating meaningful but less extreme variation compared to GDP per capita. The standard deviation (7.71) shows moderate dispersion.

Summary table

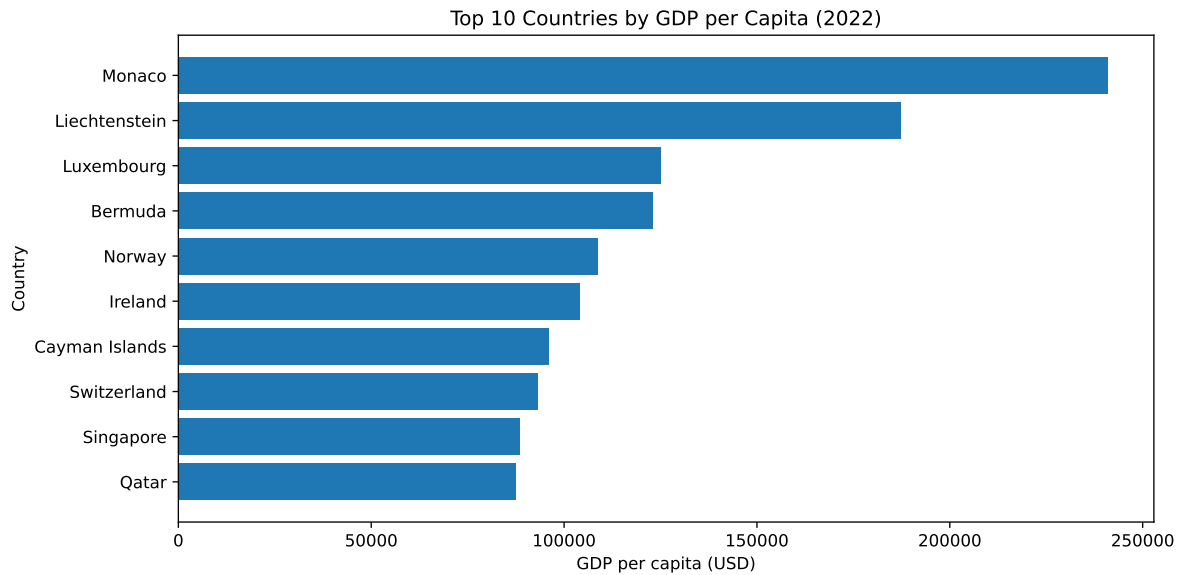


Figure 1: Top 10 Countries by GDP per Capita (2022)

```
import numpy as np

def summarize_indicator(series):
    return pd.Series({
        "count": series.count(),
        "mean": series.mean(),
        "median": series.median(),
        "min": series.min(),
        "max": series.max(),
        "missing_rate": series.isna().mean()
    })

summary_table = pd.DataFrame({
    "inflation_rate": summarize_indicator(df[ind1]),
    "gdp_per_capita": summarize_indicator(df[ind2]),
    "life_expectancy": summarize_indicator(df[ind3]),
}).T

# optional: round for readability
summary_table = summary_table.round({
    "mean": 2, "median": 2, "min": 2, "max": 2, "missing_rate": 3
})
```

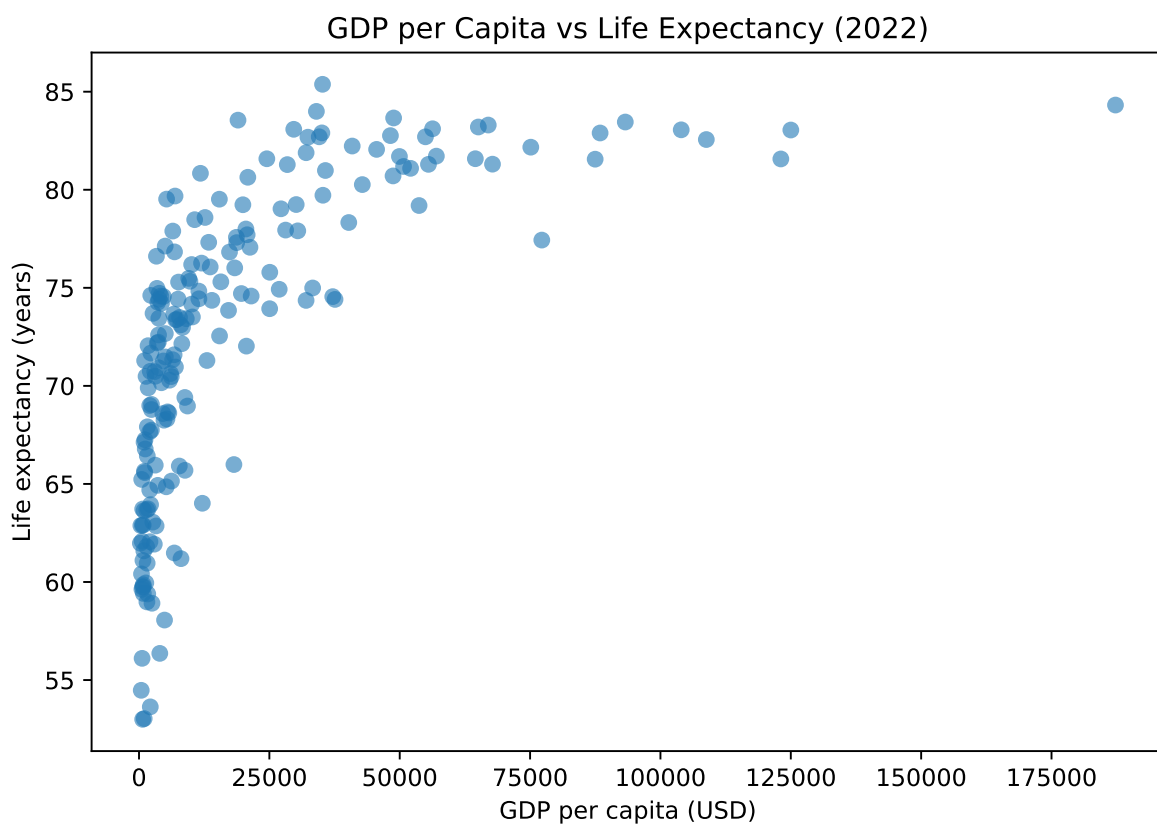


Figure 2: GDP per Capita vs Life Expectancy (2022)

Table 3

Table: Key summary statistics for the three selected indicators (2022). Source: World Bank WDI dataset.

summary_table

	count	mean	median	min	max	missing_rate
inflation_rate	169.0	12.49	7.97	-6.69	171.21	0.221
gdp_per_capita	203.0	20345.71	7587.59	259.03	240862.18	0.065
life_expectancy	209.0	72.42	73.51	53.00	85.38	0.037

Results

As shown in Figure 1, high-income countries such as Monaco and Luxembourg dominate the ranking.

The relationship between GDP per capita and life expectancy is illustrated in Figure 2, showing a strong positive association.

Key descriptive statistics are summarised in Table 3.

The data used in this analysis are sourced from the World Development Indicators dataset Bank (2022a).

Indicator definitions and metadata are provided by the World Bank Bank (2022b).

Bank, World. 2022a. “World Development Indicators.” Dataset. <https://databank.worldbank.org/source/world-development-indicators>.

———. 2022b. “World Development Indicators Metadata.” Documentation. <https://databank.worldbank.org/metadataglossary/world-development-indicators>.